

RFR Regime

(1. IBOR Transition and RFR Basics)

LIBOR – the old friend



British Bankers' Association



Deutsche Bank



JPMorganChase



Rabobank

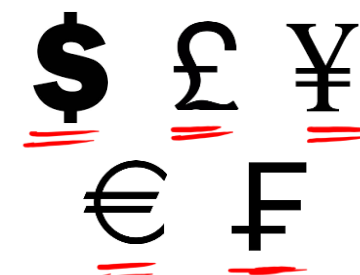


“At what rate could you borrow funds, were you to do so by asking for and then accepting interbank offers in a reasonable market size just prior to 11 a.m.?”

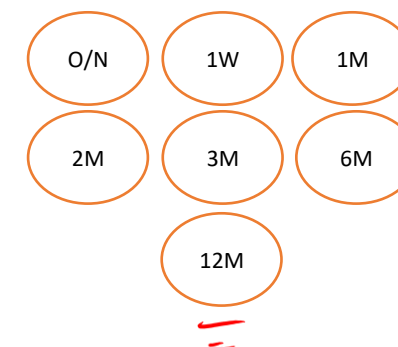
(Trimmed Mean)

LIBOR

(Currencies)

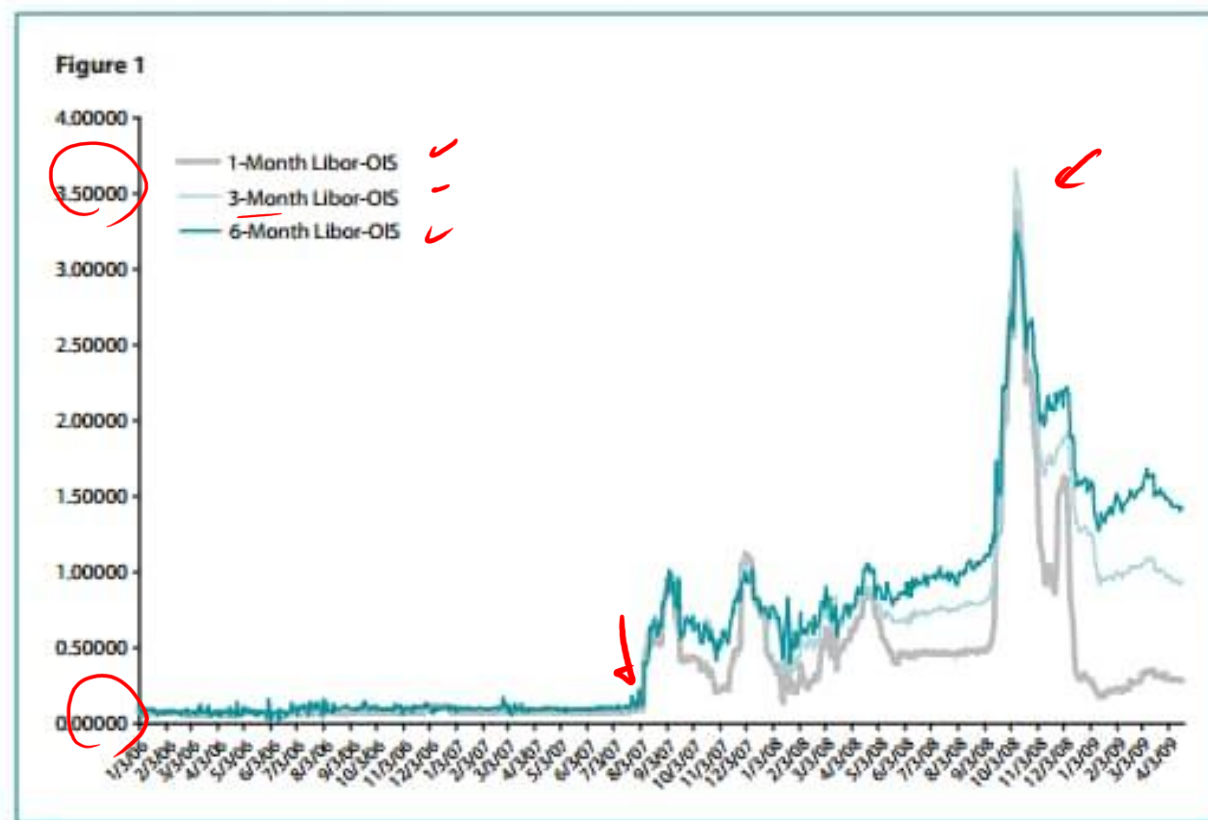


(Maturities)



LIBOR – During the 2008 Crisis

- ▶ Until 2008, LIBOR was used both as a reference curve for floating rate and risk-free curve for discounting. During the Global Financial Crisis of 2008, the spread between LIBOR and OIS skyrocketed. This led to dual-curve framework where LIBOR was used for the projected floating coupon whereas OIS was used to discount the cashflows.



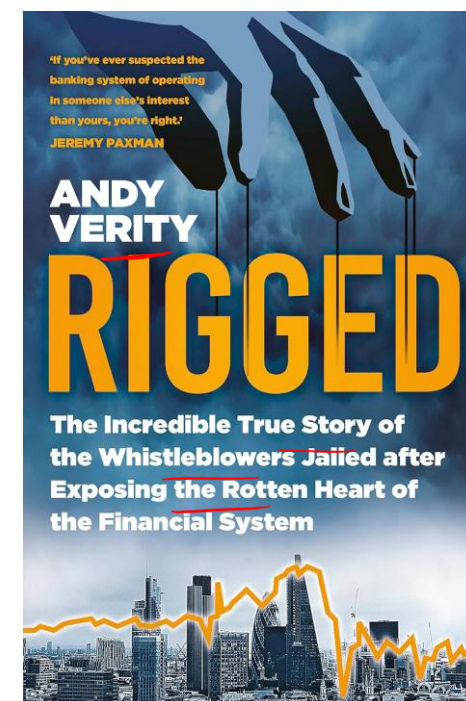
$$N \times \tau (K - \boxed{F}) P(t, T)$$

↓

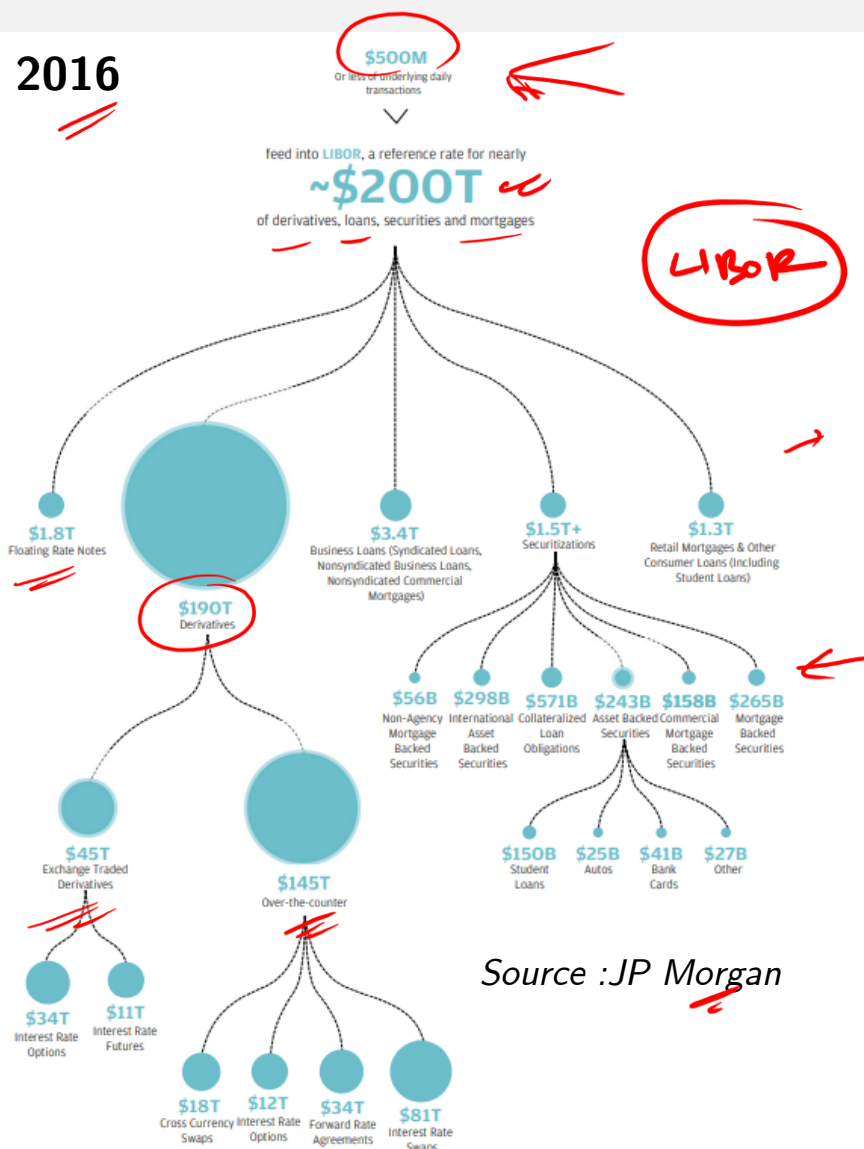
OIS

LIBOR – Breaking Bad

- ▶ In 2012, multiple criminal settlements by Barclays bank revealed fraud and collusion between member banks. It was discovered that some banks are inflating or deflating their rates to
 - ❑ Profit from their trades
 - ❑ Appear more credit-worthy than they are
- ▶ Fines totaling more than USD 9 billion were levied against large banks, including Barclays, Citigroup, Deutsche Bank, JP Morgan, Lloyds, RBS, Rabobank, and UBS



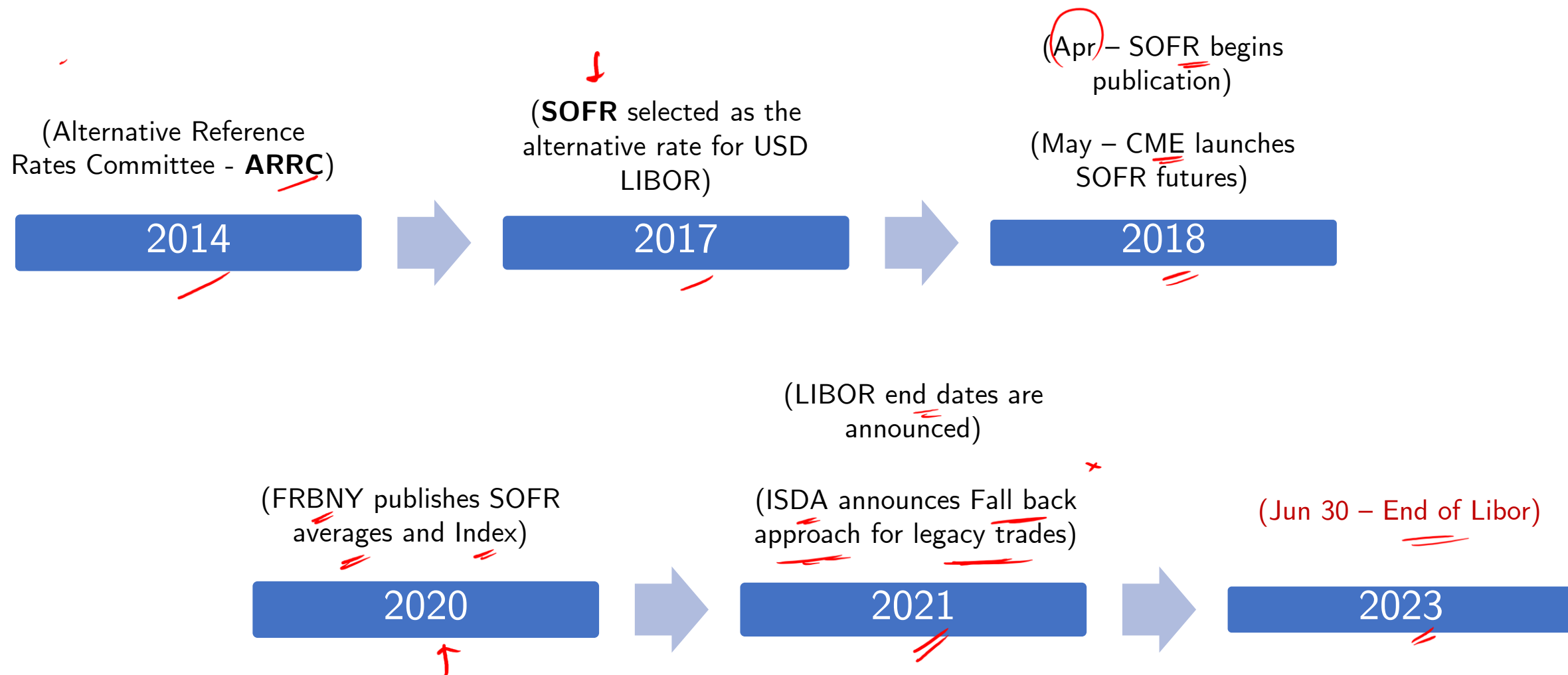
Thinning of LIBOR daily transaction volume



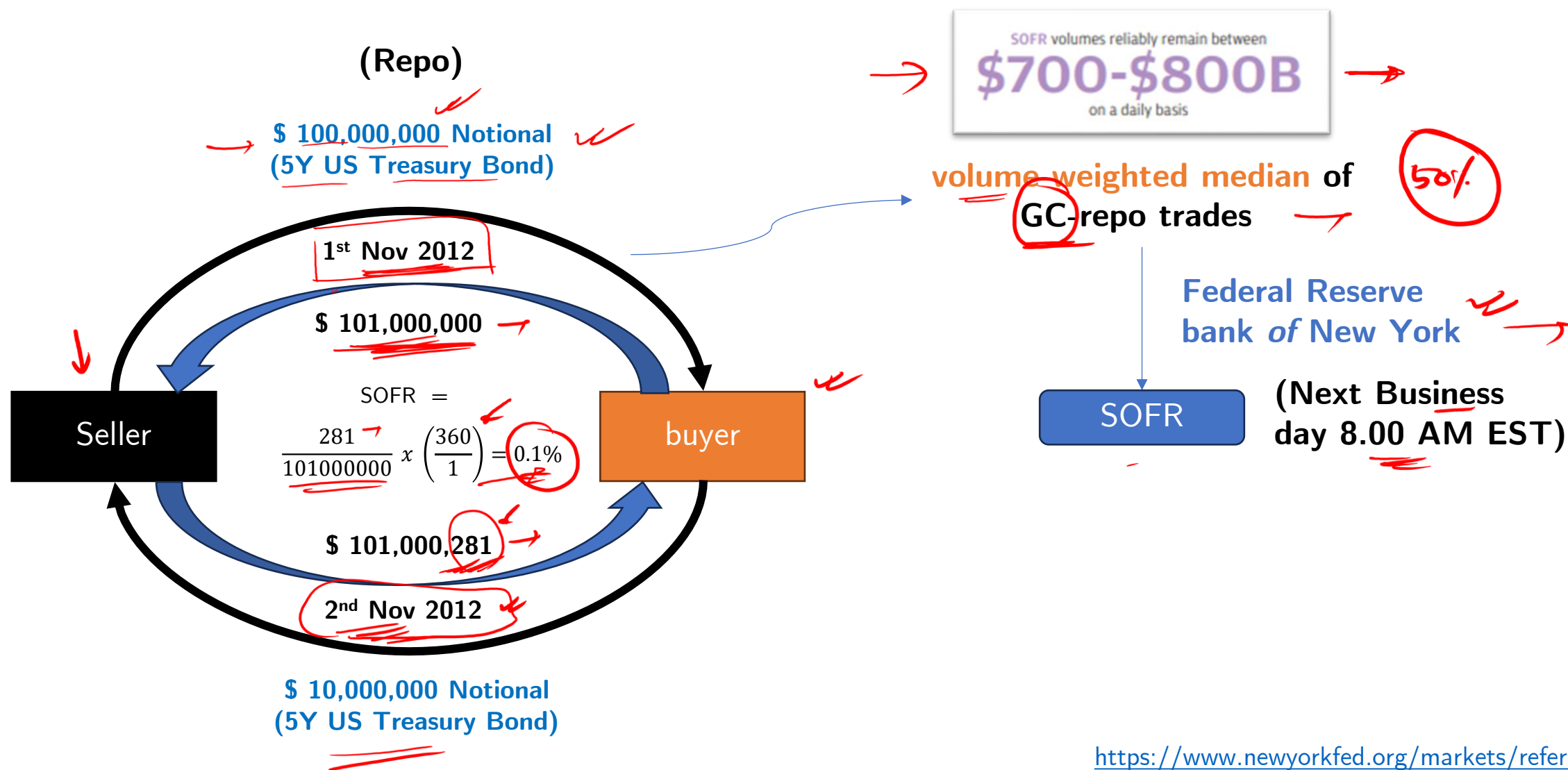
Problem with LIBOR :

\$200 trillion worth of derivatives
reference benchmark derived from **\$500 million** in average daily trading volume

LIBOR Transition



What is SOFR ?



<https://www.newyorkfed.org/markets/reference-rates/sofr>

Differences between LIBOR and SOFR

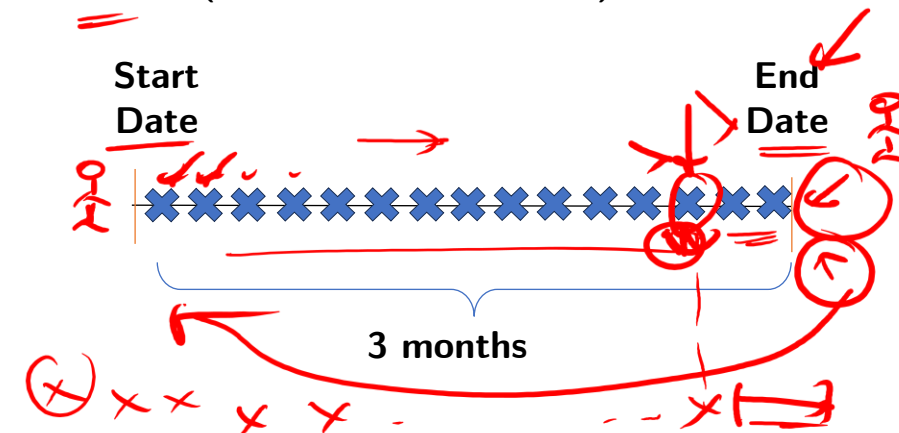
LIBOR

- Unsecured rate ✓
- Forward Looking ✓
- Various Maturities ✓
- Built-in credit component →
- Partially transaction based →
- \$500 million daily underlying transactions

3M IBOR (Determined in Advance)



3M SOFR (Determined in Arrears)



SOFR

- Secured Rate ✓
- Backward Looking ✓
- Overnight ✓
- Risk-free (RFR) →
- Wholly transaction based →
- \$750 billion daily underlying transactions

RFRs (ARRs)



Currencies	RFR	RFR Administrator	Secured
 ✓	<u>SONIA</u> (reformed)	Bank of England	No
 ✓	<u>SOFR</u>	<u>Federal Reserve Bank of New York</u>	Yes
 ✓	<u>ESTR</u>	<u>European Central Bank</u>	No
 ✓	<u>SARON</u>	SIX Swiss Exchange	Yes
 ✓	<u>TONAR</u>	Bank of Japan	No

IBOR Fallback

- ▶ For legacy IBOR derivatives which have not matured, ISDA produced their final proposal which entails using a spread (median of produced historical spread data) to arrive at the IBOR fallback rate calculated separately for each tenor.
- ▶ On 5th March 2021, the spread adjustments were fixed for all LIBOR tenors and currencies. Bloomberg is the official vendor of the fallback rates.



3m Libor? → 3m Safe rate + (S)

LIBOR	Tenor	Old Ticker (to be deprecated)	New Ticker	Spread Adjustment (%)
USD	Overnight	SUS00ON Index	VUS00ON Index	0.00644
USD	1 Week	SUS0001W Index	VUS0001W Index	0.03839
USD	1 Month	SUS0001M Index	VUS0001M Index	0.11448
USD	2 Months	SUS0002M Index	VUS0002M Index	0.18456
USD	3 Months	SUS0003M Index	VUS0003M Index	0.26161
USD	6 Months	SUS0006M Index	VUS0006M Index	0.42826
USD	12 Months	SUS0012M Index	VUS001Y Index	0.71513

Source :Bloomberg

$$r_G = \left(\prod_{i=1}^{N_b} (1 + r_i \tau_i) - 1 \right) / \tau_c = \left(\frac{\text{Index}_t}{\text{Index}_{t_0}} - 1 \right) \times \left(\frac{360}{d_c} \right)$$

11

ESTR Rates, Index & Averages

$$\text{Index}_i = \begin{cases} 100, i = \text{Oct 1st, 2019} \\ \text{Index}_{i-1} \cdot (1 + r_i \tau_i), i > \text{Oct 1st, 2019} \end{cases}$$

$$r_G = \left(\prod_{i=1}^{N_b} (1 + r_i \tau_i) - 1 \right) / \tau_c = \left(\frac{\text{Index}_t}{\text{Index}_{t_0}} - 1 \right) \times \left(\frac{360}{d_c} \right)$$

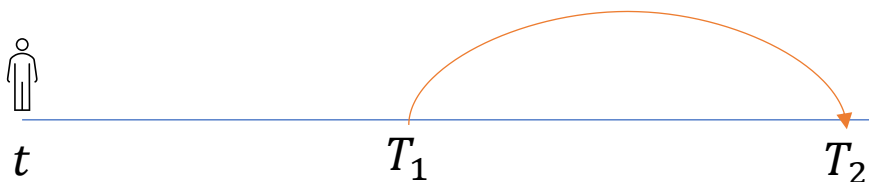
	A	B	C	D	E	F	G	H	I	J
1										
2										
3										
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8										
9										
132										
133										
134										
135										
136										
137										
138										
139										

ESTR Rates									
ESTR VALUE DATE	ESTR PUBLICATION DATE	ESTR	VALUE DAY	di	INDEX	1M	3M	6M	
2019-09-30	2019-10-01				100.000000				
2019-10-01	2019-10-02	-0.549%	Tuesday	1	99.99847500				
2019-10-02	2019-10-03	-0.551%	Wednesday	1	99.99694447				
2019-10-03	2019-10-04	-0.555%	Thursday	1	99.99540285				
2020-03-27	2020-03-30	-0.530%	Friday	3	99.72893084				
2020-03-30	2020-03-31	-0.528%	Monday	1	99.72746815				
2020-03-31	2020-04-01	-0.522%	Tuesday	1	99.72602210				
2020-04-01	2020-04-02	-0.527%	Wednesday	1	99.72456222	-0.53353%	-0.53615%	-0.53885%	
2020-04-02	2020-04-03	-0.534%	Thursday	1	99.72308298	-0.533270%	-0.536090%	-0.538760%	
2020-04-03	2020-04-06	-0.539%	Friday	3	99.71860375	-0.533080%	-0.536160%	-0.538660%	
2020-04-06	2020-04-07	-0.533%	Monday	1	99.71712736	-0.533070%	-0.536090%	-0.538410%	
2020-04-07	2020-04-08	-0.533%	Tuesdav	1	99.71565099	-0.533060%	-0.536020%	-0.538300%	

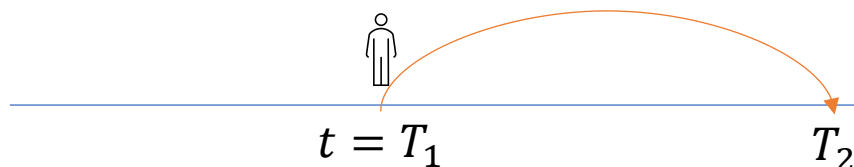
Note :

1. SOFR averages use the exact calendar days (e.g., 30, 90 etc.... for averages)
2. ESTR averages use month lags and modified business day convention if start date falls on non-business days.

Rate evolution under backward looking regime

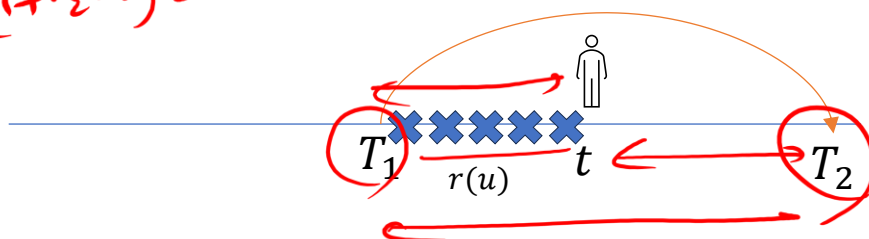


$$F(t, T_1, T_2) = \frac{1}{\tau(T_1, T_2)} \left[\frac{P(t, T_1)}{P(t, T_2)} - 1 \right]$$



$$F(t, T_1, T_2) = \frac{1}{\tau(T_1, T_2)} \left[\frac{1}{P(t, T_2)} - 1 \right]$$

$$(1 + r_1 \tau_1) (1 + r_2 \tau_2) \dots$$



$$F(t, T_1, T_2) = \frac{1}{\tau(T_1, T_2)} \left[\exp \left(\int_{T_1}^t r(u) du \right) \frac{1}{P(t, T_2)} - 1 \right]$$

Handwritten notes:
 - Red circle around $\frac{1}{\tau(T_1, T_2)}$ with arrow pointing to "annihilation factor".
 - Red circle around $\frac{1}{P(t, T_2)}$.
 - Red underline under the entire expression.
 - Red "30/360" written below the first fraction.